SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I NIKITHAA B (196526020) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	1.8	1.8	1.8
Concept Mapping Structure	25%	2.25	2.25	2.25
Linking & Relationships	20%	1.8	1.8	1.8
Application & Analysis	20%	1.8	1.8	1.8
Explanation & Clarity	15%	1.35	1.35	1.35

Total Marks 27 **/30**

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission

3 Given the concept map:

DataLink →MACAddress →LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

COT-2

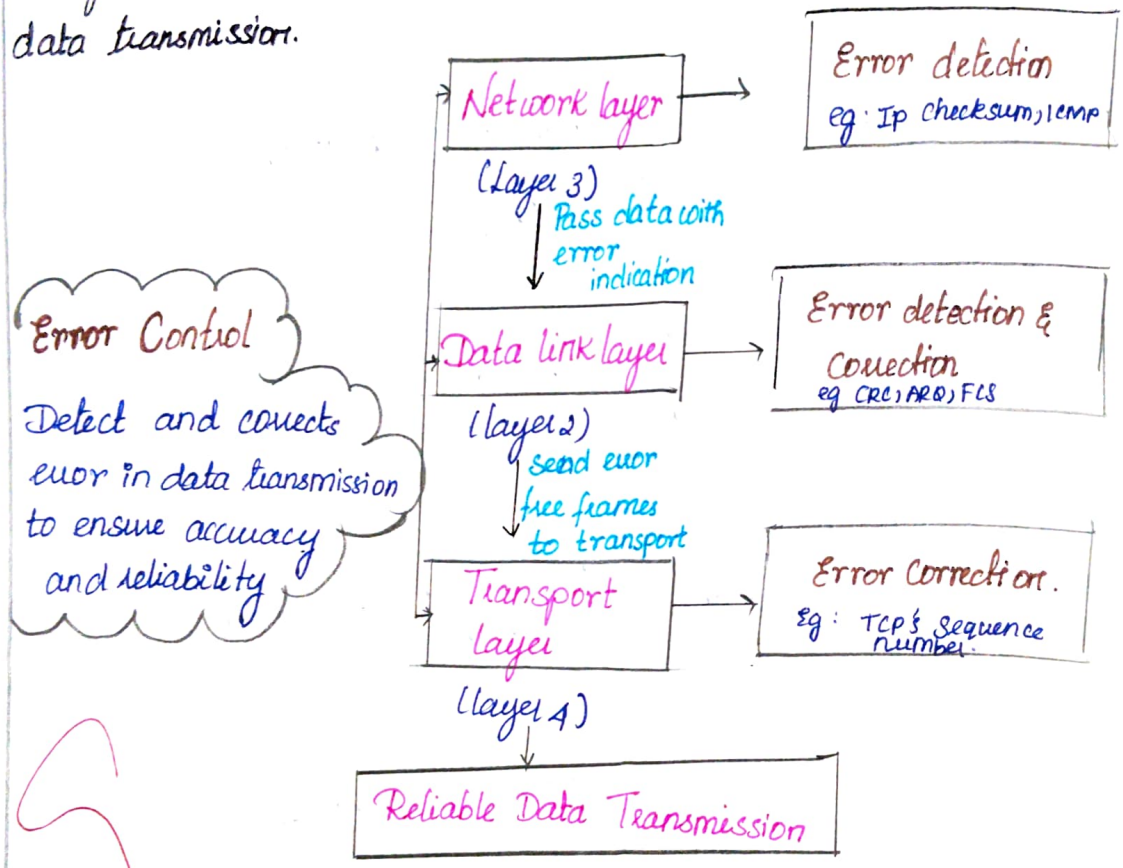
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Concept map shows:

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→ Data link layer → Errordetection → Transport layer → Error Correction.

Analyse how the interaction between these layers improves reliability in data transmission.



Network layer:-

* uses IP checksum and icmp
* Detects errors in packets
* Handles routing and delivery issues.

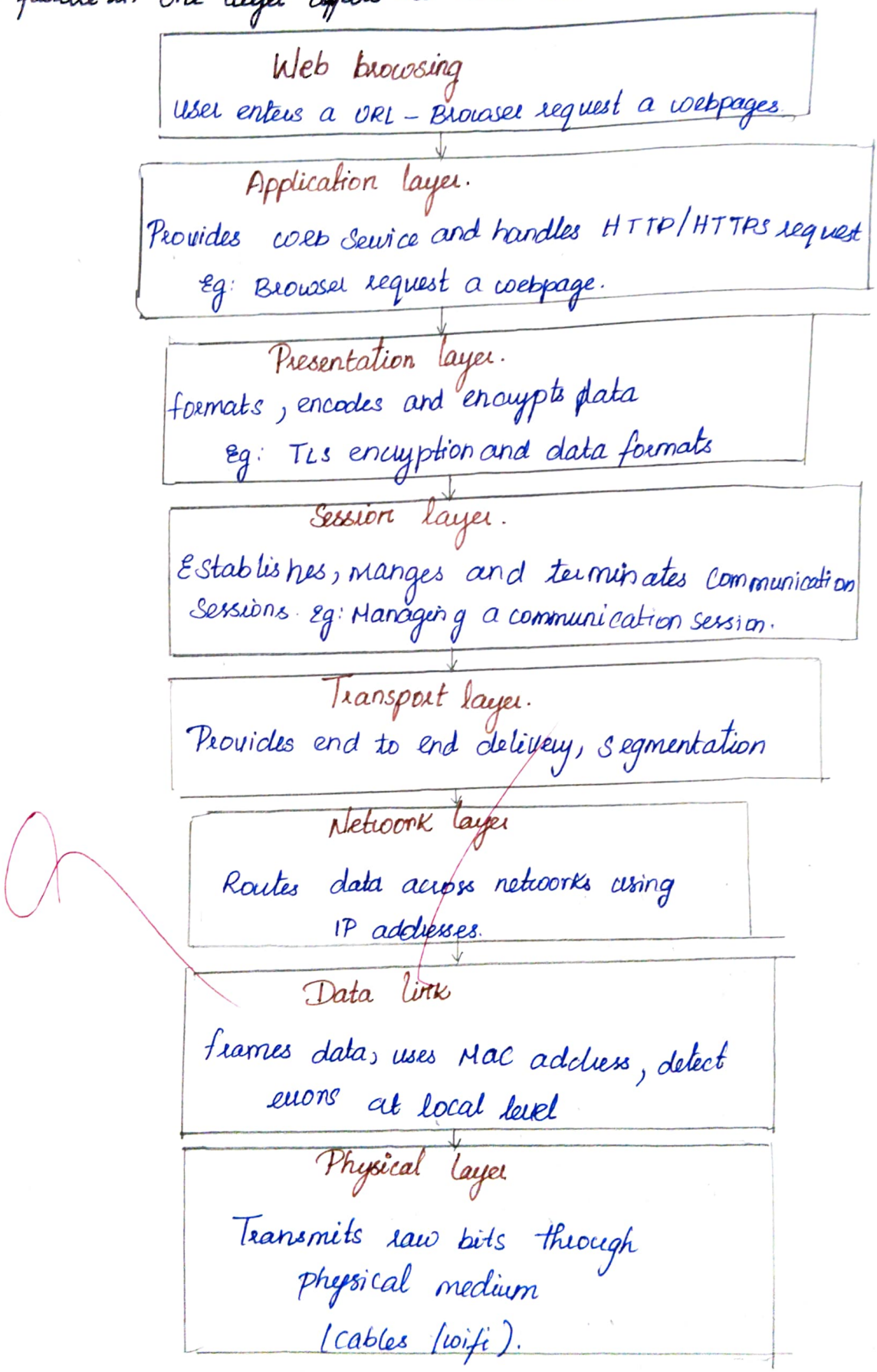
Data link layer:-

* uses CRC (cyclic Redundancy check)
* Detects and corrects frame error
* uses ARQ (Automatic Repeat request)

Transport layer:-

* uses Sequence number and ACKs
* Detect lost data
* Retransmits data for end to end reliability

5) Develop a concept map for web browsing using OSI layer and analyze how failure in one layer affects the entire Communication Process.

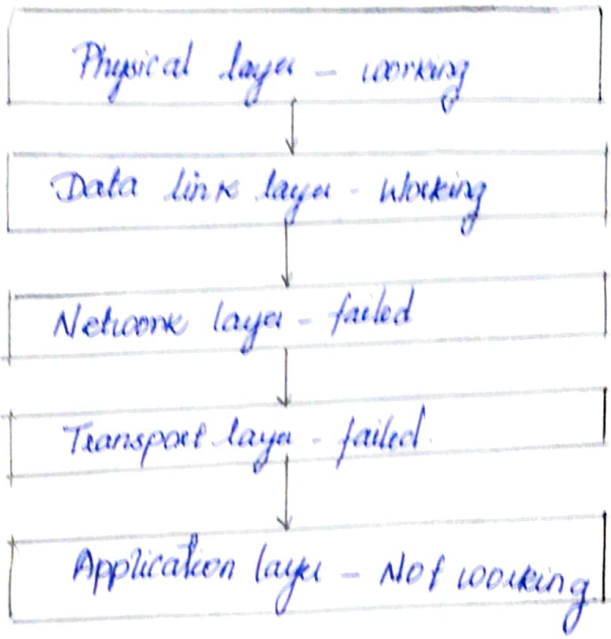


network issue is represented as the Concept Map:

Physical ✓ → Data link ✓ → Network x → Transport x → Application x

Analyze the problem and explain how the Concept Map

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Analysis:

The Concept map showing that physical and data link are working correctly but the network layer and transport layer has failed. Therefore, the problem may be related to an incorrect IP address, Subnet mask, default gateway or routing configuration. Since the network layer is not working, the transport and application layer may also fail to communicate.

Concept map help in troubleshooting

1) Identifies the problem: It highlights the network layer as the first failed layer.

2) Saves time: It avoids the checking the already working physical and data link layer unnecessarily

3) Guides testing : The technicians can check IP configuations using ip config and test Connectivity using Ping.

4) find the root cause : checking IP addresses, gateways and routing tables locate the network fault.

5) Confirms the Solution : After fixing the network layer, the technicians tests the transport and application layers again.